

BUSINESS DATA MANAGEMENT

CAPSTONE PROJECT MID TERM



Data-Driven Sales, Inventory, and Profit Optimization in Agricultural Retail Business

Submitted by:-

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Executive Summary:

This study is rooted in original data collected from collected from Krishi Khad Bhandar, it is a family-run agricultural retail store in Belhariya (Sarwan, Gaya, Bihar) supplying fertilizers, seeds, and pesticides to farmers, dealers and retailers across nearby rural areas. Over two decades the business has accumulated transaction level sales and inventory records, but these data remain underutilized. The objective of this capstone project is to analyze historical sales and purchase records to improve demand forecasting , optimize inventory levels, and recommend pricing and product mix strategy that enhance profitability while reducing stockouts and carrying costs.

This mid-term submission includes source and proof of originality, a metadata catalogue for every dataset column, descriptive statistics that summarize central tendencies, sales pattern and variability across SKUs, with mean, range, standard deviation, and skewness providing valuable insights. For instance, we find visualizations showing monthly and seasonal sales pattern, top-performing products and regions, and price demand relationship. The analysis approach combines Excel-based cleaning and pivot analyses for business-facing summaries and relationship testing. Expected outcomes are accurate short term sales forecasts for key SKUs, an ABC classification of inventory to prioritize stock control and actionable recommendations including reorder points, safety stock guidance and pricing suggestions to improve margins and reduce stockouts. The importance of category-specific strategies, such as targeted promotions, improved stock management and leveraging agri-product for revenue.

By integrating these insights, Mr. Ajesh Kumar can enhance inventory efficiency, drive sales through targeted campaigns, and better align the store's offerings with customer demand. This actionable analysis supports the store's growth while addressing its operational challenges.

Proof of Originality:

Details:

- Shop Name : Krishi Khad Bhandar
- Type of Organization : B2B and B2C both
- Shop's Address : Belhariya, Sarwan Bazar, Barachatti, Gaya, Bihar
- Shop Owner's Name : Mr. Ajesh Kumar
- Owner's Phone Number : 9955627145
- Number Of Employee : 2
- Working Time : 6:00 AM to 7:00 PM (13 hours)



Image 1&2 : Krishi khad bhandar, Gaya

Proof Links:

- Interview Video : [Chandan interview with shop owner.mp4](#)
- Video of Shop : [Shop inside video](#)
- Some more Photograph : [photo 1.jpg](#)
- Some more Photograph : [Photo 2.jpg](#)
- Some more Photograph : [Photo 3.jpg](#)

- Authorization Letter : [Authorization letter.jpg](#)

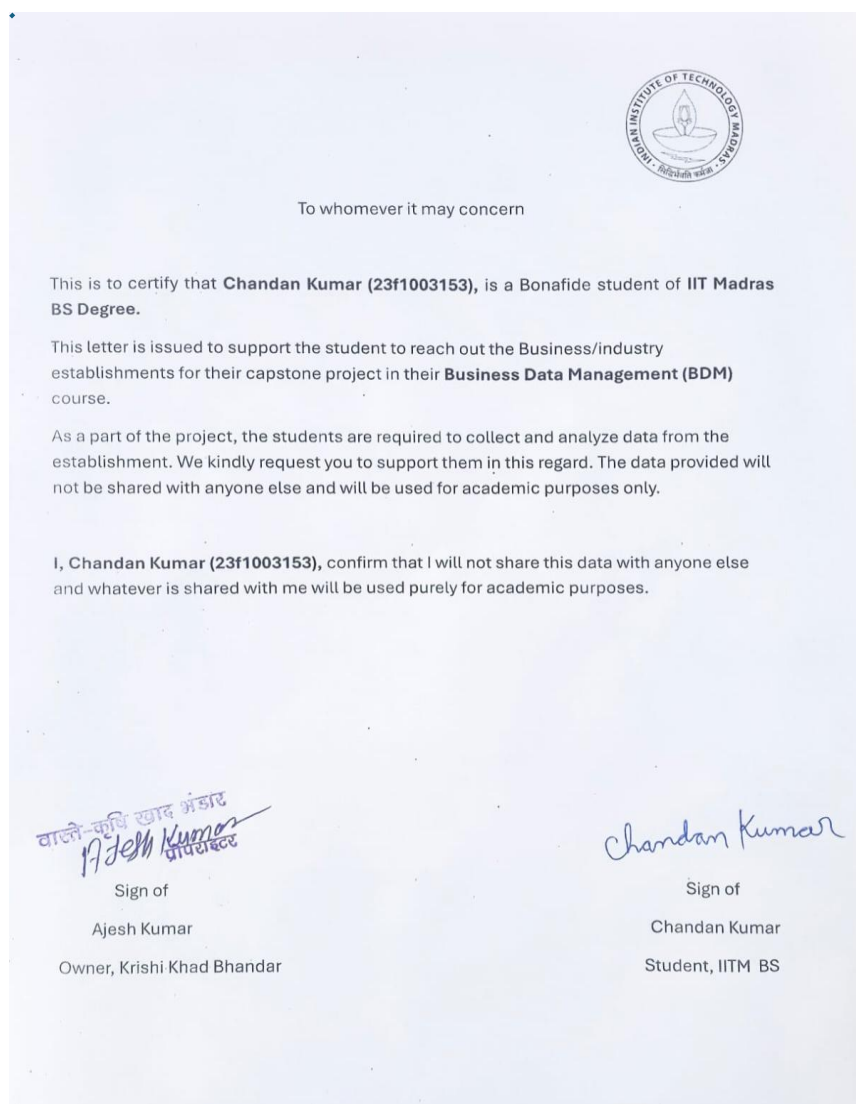


Image 3 : Authorization from Mr. Ajesh Kumar to use data for BDM Capstone Project

Meta Data:

The sales data was initially received in raw format, either in handwritten or system generated, which required digitization. Through this process, a total of 1198 bills spanning a 3 years period were compiled. To streamline the analysis, the digitized data was further consolidated, in first sheet there is raw data, in second sheet there is cleaned data and from cleaned data I categorized data in further in three sheet that is Fertilizer sheet, Pesticides sheet and Seeds sheet. Except raw data sheet, I have analyze each sheet with descriptive analysis, so analysis has done very well. This consolidation provided a clear summary of

total sales for each SKU, making the dataset more manageable and suitable for further analysis.

- **Clean Dataset Link :** [BDM project Data](#)
- Data Collection duration : 3 years
- Data Collection dates : 01-01-2023 to 12-12-2025
- Dataset Dimensions in cleaned data sheet : The dataset contains **8 columns** and **1041 rows**

date	Region	Product	Category	Weight	Price per	Quantity	Total Value
1/2/2023	Sobh	Cropyte	Pesticides	50 g	115	100	11500
1/16/2023	Bhadya	antracol	Pesticides	1 kg	535	2	1070
8/8/2023	Paroriya	IFFCO DAP	Fertilizer	50 kg	1260	10	12600
9/8/2023	Jamuhar	Cropyte	Pesticides	50 g	115	24	2760
9/8/2023	Jamuhar	Action	Pesticides	25 g	62	20	1240
9/8/2023	Jamuhar	Sticker	Pesticides	100 g	80	12	960
9/8/2023	Barachatti	coriander	Seeds	1 kg	220	10	2200
9/8/2023	Jamuhar	Mahapow	Fertilizer	1 kg	130	10	1300
9/8/2023	Barachatti	Palak	Seeds	1 kg	110	10	1100
9/8/2023	Jamuhar	Mahajivan	Fertilizer	1 kg	90	10	900
9/8/2023	Jamuhar	Mahakal	Pesticides	20 g	80	5	400
9/8/2023	Barachatti	Coriander	Seeds	1 kg	300	2	600
9/11/2023	Jamuhar	cropyte	Pesticides	50 g	115	20	2300
9/11/2023	Jamuhar	action	Pesticides	25 g	62	10	620
9/11/2023	Jamuhar	Action	Pesticides	100 g	230	5	1150
9/18/2023	Piparsot	Wheat	Seeds	50 kg	900	8	7200
9/26/2023	Barachatti	Urea	Fertilizer	50 kg	440	3	1320
9/27/2023	Piparsot	urea	Fertilizer	50 kg	440	10	4400
10/7/2023	Sobh	Ajit Mustai	Seeds	1 kg	220	2	440

Image 4 : A snapshot of cleaned data sheet how the data looks like

Columns :

1. Date : The date for which entries are maintained
2. Region : The region column show about which area are from
3. Product : The product column show about name of product
4. Category : The Category column show about which type of product are
5. Weight per bag : This column show about what is the weight of unit quantity/bag
6. Price per bag : This column show about price of unit quantity/bag

7. Quantity : Quantity column show about how many unit are purchased
8. Total Value : This column show about total price of whole quantity that product

Descriptive Statistics :

Table 1 : Descriptive Statistics Summary

statistics	Fertilizer				Pesticides			Seeds			
	Price per bag	Price per kg	Quantity	Total Value	Price per bag	Quantity	Total value	Price per bag	Price per kg	Quantity	Total Value
Mean	919.37	26.74	10.28	6858.41	175.71	10.63	1185.8	980.31	277.11	13.18	6624.8
Standard error	23.79	1.63	0.53	357.88	16.80	1.65	234.40	67.22	34.28	0.89	377.86
Median	1000	23	5	4000	105	5	550	377.5	76.5	6	4600
Mode	1200	24	5	1200	48	5	240	300	250	5	7500
Standard deviation	523.43	35.85	11.69	7873.5	186.34	18.27	2599.6	1400.48	714.24	18.50	7871.8
Sample variance	273980	1285.1	136.85	61992081	34723	333.92	6758114	1961339	510146	342.39	61965818
Kurtosis	-1.41	46.97	4.36	13.54	2.99	15.09	35.11	19.36	39.02	36.85	13.48
Skewness	0.055	5.88	2.04	3.11	1.81	3.84	5.50	4.04	5.89	4.78	3.06
Range	1865	423.6	59	63910	875	99	21490	10175	6693.2	209	59950
Minimum	35	6.4	1	90	15	1	70	25	6.8	1	50
Maximum	1900	430	60	64000	890	100	21560	10200	6700	210	60000
Sum	444975	12942	4975	3319470	21613	1308	145859	425454	120267	5720.5	2875191
Count	484	484	484	484	123	123	123	434	434	434	434

1. Fertilizer

- Price per Bag :** The mean price per bag is 919.37, while the median is 1000 indicating a slightly left-skewed distribution. Skewness(0.054) is close to zero, confirming the distribution is nearly symmetric. Kurtosis (-1.41) indicates a flatter distribution with lighter tails. The large number of 1865 (35-1900) and high standard deviation of 523.43 show high variability in bag prices. Price per bag varies widely but is fairly symmetrically distributed around 1000.
- Price per kg :** The mean price per kg is 26.74, with median is 23, showing a slightly right skew. The high skewness (5.88) and very high kurtosis (46.97) indicate extreme outliers and a highly long-tailed distribution. The maximum price (430) is significantly higher than most values, confirming outliers. Standard deviation (35.85) is higher than the mean, indicating high dispersion. Price per kg has strong right-skewness and heavily tails due to a few extremely high-priced items.

- **Quantity :** Average quantity purchased is 10.28 units, while the median is 5, showing right skew. Skewness (2.04) confirms a right-skewed distribution where most values are small but a few purchases are very large. The range (1-60) and standard deviation (11.69) indicate large variability. Kurtosis (4.36) suggests moderately heavy tails, indicating occasional large orders. Quantity distribution is dominated by small orders with a few significantly large ones.
- **Total Value :** The mean total value is 6858, but median (4000) is much lower, suggesting right skew. Skewness (3.11) and kurtosis (13.53) confirm extreme right-skewness and heavy tails. Very large values such as 64000 and 53900 inflate the mean and increase variability. Standard deviation (7873.50) is higher than the mean, showing high spread. Total values is highly influenced by small number of very large transactions.

2. Pesticides

- **Price per bag :** The mean price per bag is 175.72, but median is 105, showing that the distribution is right-skewed. Skewness (1.81) confirming a long tail due to some very expensive pesticides. Kurtosis (2.99) indicates a sharper peaks and heavier tails. The Wide range (15 to 890) and high standard deviation (186.34) indicate very high variability in unit prices. A few high-priced pesticides (e.g. 850 and 890) raise the mean significantly. Price per bag varies widely, with many low-priced products but few very high-priced ones causing right-skewness.
- **Quantity :** The mean quantity is 10.63, but the median is 5, again indicating strong right skew. Skewness (3.84) and kurtosis (15.09) show that the distribution has extreme outliers with very large purchase quantities. The maximum quantity is 100, while most values are closer to the minimum of 1-5 units. Standard deviation (18.27) is high relative to the mean, showing large variability. Quantity distribution shows many small purchases but a few extremely large purchases, creating a heavily skewed pattern.
- **Total value :** The mean total value is 1185.85, but the median is only 550, indicating strong right skew. Skewness (5.50) is extremely high, showing the presence of very large transactions. Kurtosis (35.11) indicates a very heavy-tailed distribution (extreme outliers). The range is huge (70 to 21560) confirming the presence of high value invoices. Standard deviation (2599.63) is more than double the mean , highlighting high volatility. Total value has extreme right skewness and heavy tails due to a few very large pesticides.

3. Seeds

- **Price per bag :** The mean price (980.31) is much higher than the median (377.5), indicating a highly right-skewed distribution. Skewness (4.04) conforming strong positive skew due to expensive premium seed varieties. Kurtosis (19.36) shows very heavy tails , meaning extremely large price outliers. Range (25 to 10200) which is extremely wide. High standard deviation (1400.47) further confirms variability.
- **Price per kg :** Mean weight per kg (277.11), while median (76.5), again showing heavy right skew. Skewness (5.89) and kurtosis (39.03) indicate very strong positive skew due to extremely high per-kg prices for certain seed types. Extremely high shows long tails and many extreme points. A small number of very high-priced items distort the distribution. Standard Deviation (18.50) high variability occasional bulk purchases increase spread. Quantity purchased is dominated by small orders, but handful of large bulk purchases create extreme right-tailed behavior.
- **Total Value :** Mean (6624.86) is much higher than median (4600). Skewness (3.06) and kurtosis (13.48) strong right skew caused by very high value transactions. Standard Deviation (7871.84) very high spread indicating transactions vary from very small to extremely large. Range (50 to 60000) very wide range, indicating some very large purchases. Total revenue per transaction varies widely, although most sales cluster around 3000-4000, large bulk purchases drive the mean upward.

Methods of Analysis with Justification :

1. Data Cleaning and validation

- **Process :** The Shop's sales data was maintained in handwritten records. Over multiple visits, this data was digitized into Excel to facilitate analysis. During this process, issues like missing values (name of product, quantity, cost), duplicate entries and discrepancies in SKU names were identified and resolved. After cleaning, the data was consolidated to summarize daily sales for each category.
- **Justification :** Cleaning ensures accuracy and consistency, while consolidation provides a holistic view of sales trends. Accurate forecasts and inventory recommendations require clean transactional data. Consolidation also makes it easier to analyze long-term trends and make data-driven decisions.

2. Descriptive Statistics Analysis

- **Process :** Key statistical metrics were calculated for each category, such as mean, median, mode, range, standard deviation, skewness, and kurtosis. These metrics were used to summarize the performance of each category. This shows central tendency and dispersion metrics for reporting and for confirming normality or skewness.
- **Justification :** This step provides a detailed understanding of each category's behavior. For categories like fertilizer, seeds and pesticides, steady demand patterns were identified, ensuring optimal stock levels without overstocking. For discretionary items like paddy and wheat, high variability highlighted the need for dynamic stocking strategies to meet peak demand during monsoon and winter respectively. Descriptive statistics allow to prioritize essential items and manage shelf space effectively.

3. Trend Analysis

- **Process :** Sales data was analyzed over time to identify seasonal patterns, monthly variations, and monsoon-related spikes. For category-wise analysis like fertilizer, the trend shows strong seasonality, with two main peaks (Kharif & Rabi). Urea and DAP contribute to sudden volume spikes during the beginning of cropping cycles.
- **Justification :** Trend analysis reveals that sales at Krishi Khad Bhandar closely follow agricultural cycles. Fertilizers drive volume during sowing months, seeds show strict seasonal behavior, and pesticides generate revenue spikes during pest incidents. These insights are crucial for optimizing stock levels and planning procurement throughout the year.

4. ABC Segmentation analysis

- **Process :** A-category products dominate revenue, mainly fertilizers and costly pesticides. These should always be well stocked. B-category items are seasonal and support steady business. They require balance between availability and storage cost. C-category items add variety but do not significantly impact profits. Overstocking

them leads to capital blockage. Prioritizing procurement and forecasting efforts on A-items will produce the greatest impact on business performance.

- **Justification** : ABC Segmentation provided valuable insights into which products contribute the most to revenue. Fertilizers dominate A-category, pesticides fluctuate between A and B categories, seeds commonly fall under B and C categories. This analysis helps Krishi khad Bhandar optimize purchasing, reduce waste, and ensure that the capital is invested in high-impact items.

5. Correlation analysis

- **Process** : Correlation analysis helps determine how two numerical variables are related to each other. Quantity sold is the strongest influencer of Total sales, especially for fertilizers. Unit Price plays a smaller role in determining quantity sold. Pesticides show moderate price-driven revenue behavior, unlike fertilizers and seeds. Demand is mostly seasonal and need-based, not price driven. Strong correlation patterns help identify which items should be stocked heavily and which require pricing strategy adjustments.
- **Justification** : Correlation analysis shows that revenue is primarily driven by quantity for fertilizers and seeds, while unit price drives revenue for pesticides. The overall demand is only weakly influenced by price, confirming that agriculture inputs are essentials goods with predictable consumption patterns.

Results and Findings:

The line chart illustrates the month-wise quantity sold for three major product categories fertilizers, pesticides, seeds across the calendar year. The objective is to understand seasonal demand patterns and identify peak sales periods for each category. Seeds demand is extremely seasonal and peaks only during sowing periods. It spikes very high in July due to high demand of paddy seeds and in November due to wheat seeds. Stocking must be done before June-July to avoid shortages. Fertilizers follow a predictable bi-seasonal pattern, and high inventory should be planned before July and again before October. Pesticides demand is

problem-driven, not seasonal, and quantities remain low compared to fertilizers and seeds.

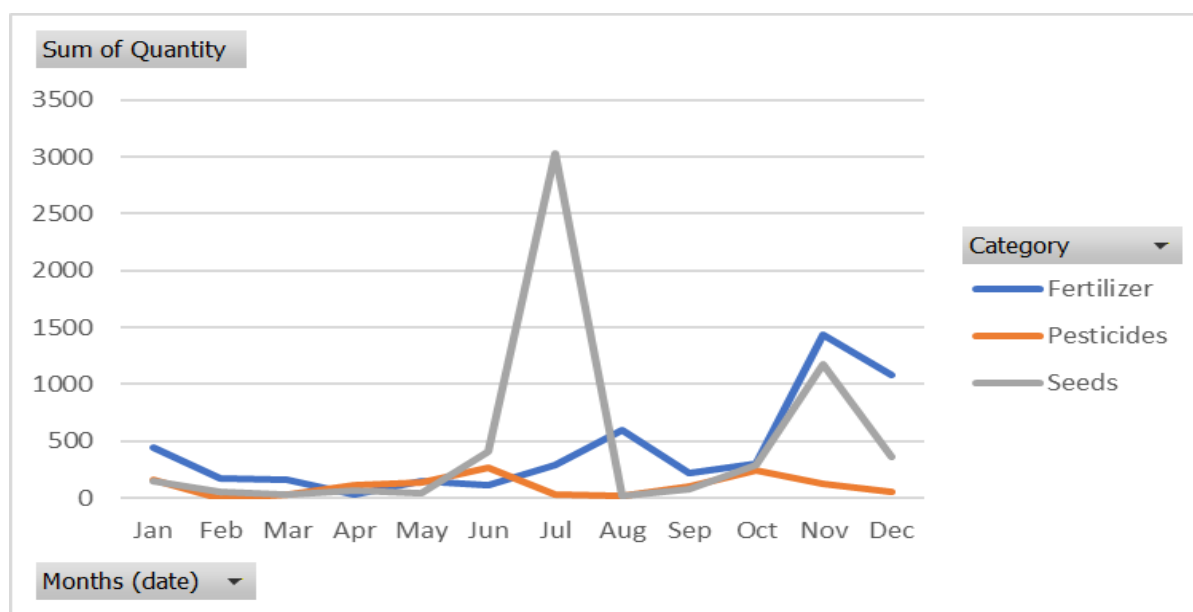


Image 5 : Line chart show sales trend of 3 categories

The result clearly show that agricultural input sales follow a predictable seasonal patterns, fertilizers dominate overall movement, pesticides generate high unit revenue, and seeds have sharp seasonal spikes. These findings provide actionable insights for inventory management, pricing strategy, forecasting and overall business planning.

	<i>Fertilizer</i>	<i>Pesticides</i>	<i>Seeds</i>
Fertilizer	1	0.177316	0.510012
Pesticides	0.177316	1	0.452052
Seeds	0.510012	0.452052	1

Image 6 : Correlation heatmap between categories

The Correlation patterns between Fertilizers, Pesticides, and seeds provide meaningful insights for agricultural input suppliers, distributors, and policy planners. Understanding how these product categories move together across villages helps in making informed decisions regarding inventory, pricing, marketing and distribution strategies. The moderate correlation especially between Fertilizers & seeds (0.51) and seeds & pesticides (0.45) indicate that demand for these categories tends to rise together in specific villages. Villages with strong seed-pesticide correlation may need pest management awareness programs. Villages with high seed-fertilizer correlation may be ideal for nutrient management campaigns.